ADVANCED PIXEL VALUE DIFFERENTIATION FOR EFFICIENT AND SECURE IMAGE STEGANOGRAPHY

J. Swarna latha
Department of Computer Science and Engineering
CMR Technical Campus, Hyderabad, Telangana, India
Swarnalatha.cse@cmrtc.ac.in

B. Krishnaveni
Department of Computer Science and Engineering
CMR Technical Campus, Hyderabad, Telangana, India
Roll No: 237R5A0506
237r5a0506@cmrtc.ac.in

B. Rishik Babu
Department of Computer Science and Engineering
CMR Technical Campus, Hyderabad, Telangana, India
Roll No: 237R5A0502
237r5a0502@cmrtc.ac.in

V. Anil Kumar
Department of Computer Science and Engineering CMR Technical Campus, Hyderabad, Telangana, India Roll No: 227R1A0564
227r1a0564@cmrtc.ac.in

B. Sekhar
Department of Computer Science and Engineering
CMR Technical Campus, Hyderabad, Telangana, India
sekharaije@gmail.com

Dr K MAHESWARI
Department of Computer Science and Engineering
CMR Technical Campus, Hyderabad, Telangana, India
mahi.kirubakaran@gmail.com

Abstract*—With the proliferation of data-oriented communication networks, there is a dire need for sensitive information protection that is transmitted on public networks. Steganography best fits as an alternative in which uncertainty on the hidden data is in image and audio, while the payload cannot statistically be differentiated. This work is aimed at proposing an improved image steganography method using ADPVD for embedding capacity enhancement without compromising either the quality of the image or the performance of concealment. In the proposed approach, as the values of pixels would be changed after embedding, it predicts ahead the variation amount of intensities for the optimal selection of blocks. This enhances the robustness against the steganalytic techniques such as visual distortion analysis and structural detection. Unlike the traditional pixel Value Differentiation approaches, static quantization range tables are used and adaptive thresholding is employed in this approach to achieve an enhanced pair-pixel selection significantly reduce the distortion. Besides, the proposed scheme outperforms the state-of-the-art PVD-based schemes in the performances of PSNR and SSIM as well as the embedding payload capacity. This also proves that the proposed ADPVD scheme can be applied for secure and effective information hiding: maintaining the invisibility property when showing with classified data in an image- based communication system [1].

Keywords *Image steganography, ADPVD, PVD, data hiding, adaptive embedding, spatial domain, imperceptibility, embedding capacity, stego-image, cover image, pixel pairing, adaptive range, visual quality, steganalysis, robustness, secure communication, PSNR, SSIM, image processing.*

INTRODUCTION

The need for secure communication has increased exponentially with the exponential growth of digital data exchange because of the internet. While cryptography protects data confidentiality by transforming data into an unintelligible format, it often exposes the existence of secret communications to probable attack and interception. Steganography is the audio, and video files. Digital images are the most used among

The need for secure communication has increased exponentially with the exponential growth of digital data exchange because of the internet. While cryptography protects data confidentiality by transforming data into an unintelligible format, it often exposes the existence of secret communications to probable attack and interception. Steganography is the method of hiding the very existence of sensitive information within ordinary-appearing digital media, including image, audio, and video files. Digital images are the most used among these media due to their high redundancy, ease of transmission, and resistance against slight changes [6].

These techniques for image steganography try to balance three main factors, which are imperceptibility, embedding capacity, and robustness. Embedding capacity tells about the amount of secret data a technique can hide, whereas imperceptibility guarantees the visual quality of a cover image, which should not change for the human eye. Robustness relates to how resistant the stego-image is to common image processing operations and modern methods of steganalysis. This trade-off between these variables is still a large research challenge in the field of information hiding [2].

Conventional spatial-domain steganography methods, such as those based on substitution of the Least Significant Bit (LSB), are simple but also provide a large capacity in payload size; however, they show low robustness against statistical attacks.

PVD methods have been developed to overcome these limitations by adaptively embedding secret data, relying on the differences between the values of neighboring pixels. With respect to preserving good visual quality, smooth regions allow only limited embedding, while high pixel variation regions allow the embedding of larger data [1],[5].

LITERATURE REVIEW

Image steganography is a field of active research, and various methods have been proposed to achieve a suitable trade-off

among embedding capacity, imperceptibility, and security. Due to their simplicity and low computational burden, most of the early approaches were performed in the spatial domain. In this context, the method of LSB substitution was widely adopted for its relatively high embedding capacity. However, it has been shown in many studies that LSB substitution is vulnerable to statistical steganalysis and also to common image processing operations, which limit its applicability in secure communication scenarios [6].

To overcome the weaknesses of LSB-based schemes, PVD methods were proposed to control embedding by exploiting differences among neighboring pixels. The studies in PVD methods have shown that the so-called edge regions in images may bear larger payloads without visible visual distortion while smooth regions usually allow only very limited changes. This adaptive nature significantly enhanced imperceptibility compared to uniform embedding approaches. Subsequent works extended the PVD framework by refining the range tables and pixel grouping strategies to enhance payload capacity and reduce distortion [2], [4],.

A number of hybrid methods have combined PVD with LSB substitution in order to exploit their respective advantages. These techniques embed LSB in select smooth areas while using PVD for the edge areas, enabling them to achieve enhanced data hiding efficiency. However, unequal payload distribution due to fixed decision rules and static threshold values used in these methods results in increased detectability by advanced steganalysis techniques [3], [7].

Transform-domain steganography methods, such as DCT- and DWT-based approaches, have also been investigated in an attempt to increase robustness against compression and filtering operations. While these methods boast much stronger resistance to attack, many of them also involve increased computational complexity with reduced embedding capacity compared to the spatial-domain approaches. Because of this, most spatial-domain techniques, especially those based on PVD, remain widely investigated for real-time and resource-constrained applications.

Recent research efforts have therefore concentrated on enhancing PVD-based schemes by using adaptive range selection, dynamic pixel pair analysis, and intelligent embedding strategies. Adaptive PVD methods avoid the drawbacks of predefined range tables by dynamically adapting the embedding parameters depending on local image characteristics. Experimental results from recent studies reveal

that such adaptive approaches can yield higher PSNR and improved security without much sacrifice in payload capacity. Although several existing methods have demonstrated promising results, the challenge is to strike an optimal balance among imperceptibility, capacity, and robustness. Therefore, advanced PVD-based steganography methods which may further refine the pixel differentiation mechanisms and embedding strategies are of great interest. The proposed work supplements the ongoing research efforts in this field with an Advanced Pixel Value Differentiation method, ADPVD, designed to yield improved data hiding performance with strong resistance against steganalysis [1], [5], [8].

PROBLEM STATEMENT

Secure transmission of sensitive information over open communication channels remains a big challenge in view of the emergence of sophisticated cyber-attacks and methods of steganalysis. While image steganography offers an efficient way of hiding secret data, many previously developed techniques cannot provide an optimal trade-off among embedding capacity, imperceptibility, and security. Of all simple spatial-domain methods, the LSB substitution is particularly susceptible to statistical analysis, thereby making it easier to detect and extract the hidden information

PVD-based steganographic methods were thus proposed to enhance imperceptibility by adapting data embedding according to local variations in pixel values. Conventional PVD methods, however, use predefined range tables and embedding rules with limited adaptability for various image characteristics. A fixed threshold usually leads to poor efficiency in payload distribution, excessive modification in some regions, and loss of image quality due to variation in image textures .

Most of the existing PVD-based and hybrid steganography methods have limited robustness against modern steganalysis methods and common image processing operations such as noise addition, filtering, and compression. Moreover, the static embedding strategies of these current techniques lead to predictability, hence increasing vulnerability. Besides, not much effort has been spent on the optimization of pixel pair selection and adaptive range determination, which directly affects both security and visual quality .

Therefore, there is a need to develop an advanced technique of image steganography that will be dynamically adaptive to local features of the image, enhance the embedding capacity, preserve high visual quality, and improve robustness against steganalysis. Addressing these limitations forms the core problem that this research aims to solve through the development of an Advanced Pixel Value Differentiation-based steganography approach,[2].

Key aims of the system are:

The design of an ADPVD-based image steganography approach wherein the embedding capacity adapts dynamically to the local variation in pixel intensities within the cover image.

- Embedding capacity can be increased without sacrificing image fidelity by optimizing pixel pair selection and adaptive range used for embedding.
- Enhancing robustness against steganalysis and common image processing attacks, such as added noise, filtering, and compression.
- Attain higher PSNR and SSIM performance metrics, among others, compared to traditional PVD-based steganography methods. - Ensuring secure and efficient data transmission for real-world image-based communication systems.

DESIGN

Overview of the Proposed System

The proposed system presents the image steganography framework based on the Advanced Pixel Value Differentiation image while maintaining high visual quality and robustness.

technique, designed to securely embed secret data in a digital The system is applied in the spatial domain and utilizes adaptive pixel analysis to determine optimal embedding regions for improved imperceptibility and embedding efficiency .

This is the set of well-defined modules in this architecture. That cooperatively act in performing secure data hiding and Extraction The general process involves the following : Acquisition of input image; pre-processing of the secret data. Adaptive analysis of pixel pairs. Embedding data using ADPVD.logic, and generating the stego-image.On the receiver side, the Reverse ADPVD provides the embedded data with high accuracy process[2].

System Architecture Components

The major components of the proposed system are described in subsequent sections:

1. Selecting Cover Image

A greyscale or colour image is used as the cover image. This image acts as a carrier of embedding the secret information. Preferably, images with sufficient variation in pixels are used to achieve higher embedding capacity with less visual distortion.

2. Secret Data Input

The secret message may be text or binary data and, in any case, is converted to a binary format. This pre-processing step makes it compatible with the embedding mechanism and allows for efficient bit-level manipulations.

3. Pixel Pair Analysis

The cover image is divided into non-overlapping pixel pairs. For every pair of pixels, the difference of pixel intensity values is computed. This difference will classify the pixel pairs into smooth or edge regions that again affect the number of bits that can be embedded.

4. Adaptive Range Determination-ADPVD Logic

Unlike traditional PVD schemes, which rely on fixed range tables, in the proposed ADPVD approach embedding ranges are determined dynamically based on the local characteristics of the pixels. This adaptive approach embeds a higher payload in the edge region and keeps smooth regions for image quality maintenance.

5. Data Embedding Module

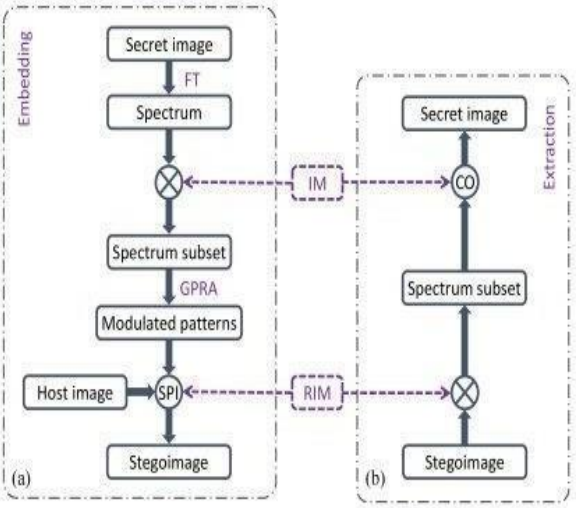
Based on adaptive range selection, the embedding of secret bits into pixel pairs will be done by changing their intensity values within the range that is permissible. Enough care will be taken to ensure pixel values fall within the valid grayscale range, avoiding any overflow or underflow.

6. Stego-Image Generation

After the embedding of all the secret data, the modified pixel pairs are combined to obtain the stego-image. This image visually looks very similar to the original cover image; therefore, no hidden data could be perceived by the human eye.

7. Data Extraction Module

At the receiving end, the stego- image is processed through the reverse ADPVD. The same analysis of pixel pairs and adaptive range logic are followed for the precise extraction of the secret hidden data with no need for the original cover image.



Flow Chart: Frequency-domain transformations in the proposed flow ensure imperceptible hiding of a secret image within a cover. A spectrum-based embedding approach results in the separation of embedding and extraction phases, hence more secure. Spectrum subset selection and modulation offer robustness against various attacks and image processing operation.

METHODOLOGY

Proposed method applies an ADPVD-based image steganography technique, which securely embeds secret information in a digital image. This approach enhances the traditional PVD by introducing adaptive range selection and optimized handling of pixel pairs, hence enhancing embedding capacity, imperceptibility, and robustness. The proposed methodology has been divided into two major phases: data embedding and data extraction.

6.2 Data Embedding Process

The embedding process conceals the secret data into the cover image through adaptive pixel analysis. Such procedures can be represented as follows:

Input Cover Image

The original cover image is then selected if required and converted into gray scale. The image is represented as a matrix of pixel intensity values.

Secret Data Preparation

The secret message is first converted to a binary bitstream. This will ensure that embedding can be efficiently carried out at the bit level.

Pixel Pair Formation

The image is divided into nonoverlapping pixel pairs, for which the absolute difference of the pixel values is calculated

.**Adaptive Range Selection**

Adaptive range, based on the computed pixel difference, is selected dynamically without using fixed predefined ranges. This step is the core enhancement of the ADPVD technique.

Bit Allocation

The adaptive range determines the number of secret bits to be embedded in each pixel pair: more bits in the edge region and no embedding in the smooth regions, which are preserved for better visual quality.

Modifying the Value of Pixels

The pixel values are adjusted to embed the secret bits, taking care that the resulting values fall within valid intensity limits of 0 through 255.

Stego-image generation

Finally, after embedding all the secret bits, the modified pixel pairs are combined again to form the final stego-image.

6.3 Data Extraction Process

The extraction process elicits the concealed data from the stego-image without relying on any original cover image.

Input Stego-Image

The input to the extraction module is the stego image. Pixel Pair Analysis Similar to the embedding step, divide the stego- image into nonoverlapping pixel pairs. Difference Calculation. The differences in pixel intensity are recalculated to determine the corresponding adaptive ranges. Bit Extraction Based on the adaptive range and pixel differences, the embedded secret bits are extracted. Secret Data Reconstruction Extracted bits are then combined to give the reconstructed original secret message. 6.4 Advantages of Proposed Methodology Adaptive embedding improves imperceptibility and security. Higher embedding capacity than in traditional PVD Resistance to steganalysis and image processing attacks Simple and computationally efficient implementation

MODULAR DESCRIPTION

7.1 Experimental Setup

An experimental evaluation has been done regarding the effectiveness of the proposed ADPVD-based image steganography technique in terms of imperceptibility, embedding capacity, and robustness. Standard grayscale test images of fixed dimensions were used as cover images for consistency in evaluation. The secret data in this experiment consisted of textual information that was converted into binary format prior to embedding.

The experiments were carried out in a controlled environment by using a software-based simulation platform. The proposed technique of ADPVD was implemented and tested against conventional PVD-based steganography methods to carry out a comparative analysis. Several test images were used, and different amounts of secret data were embedded to analyze the impact on image quality and performance metrics.

7.2 Performance Metrics

These standard metrics were used for the performance evaluation of the proposed system:

1. Peak Signal-to-Noise Ratio - PSNR

PSNR basically shows how much the stego-image deviates from the original cover image. A higher PSNR indicates better imperceptibility with less distortion.

2. **SSIM –** Structural Similarity Index SSIM computes the similarity based on luminance, contrast, and structure between the cover and stego images. Values closer to 1 indicate higher similarity.

3. Embedding Capacity

Embedding capacity refers to the quantity of secret data that can be hidden in the cover image without visibly degrading its perceptual quality.

4. Robustness

Robustness: This implies the resistance of a stego image against common image processing operations like addition of noise, filtering, and compression with no destruction of hidden data.

7.3 Results and Analysis

These results are further confirmation that the proposed approach based on ADPVD outperforms conventional schemes of PVD. The generated stego-images of the proposed approach give minimum visual distortion, even at higher embedding capacities. The adaptive range selection mechanism helps to distribute the payload efficiently over the edge and smooth regions, thereby achieving better image quality.

Also, in this case, PSNR values obtained using the proposed method are considerably higher than those obtained using conventional techniques, which clearly implies enhanced imperceptibility. At the same time, values of SSIM are maintained close to unity, establishing a strong structural similarity between cover and stego-images. Optimized pixel pair selection enhances the embedding capacity remarkably, while robustness tests confirm increased resistance to common image processing attacks.

7.4 Comparative Discussion

The proposed ADPVD technique effectively addressed the limitations of fixed range tables and static embedding rules compared with existing PVD-based methods. By nature, the proposed adaptive system enhances the security and imperceptibility without any increment in the computational complexity. These results validate the suitability of the proposed method for secure image-based data hiding applications.

RESULTS

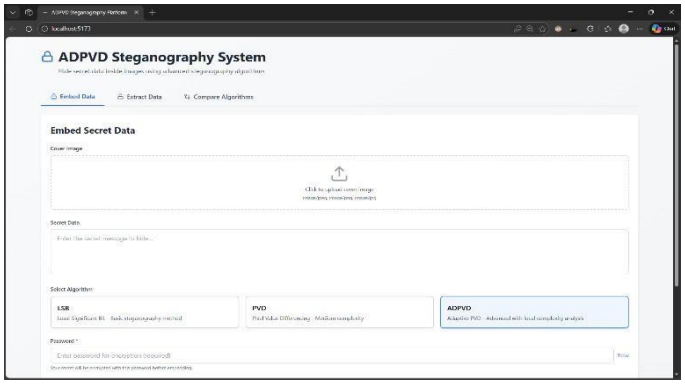


Fig 1: ADPVD Steganography System – Embed Data Interface: User interface of the proposed ADPVD Steganography System during the data embedding process. The interface provides the facility to upload a cover image, the secret message the user wants to hide, and the selection of the steganography algorithm. There are three algorithms available, namely: LSB, PVD, and ADPVD, for comparison; ADPVD is selected as the advanced option. This interface provides a password-based encryption mechanism for enhancing security before embedding. It demonstrates that this proposed system is easy to use and flexible for the purpose of secure data hiding.

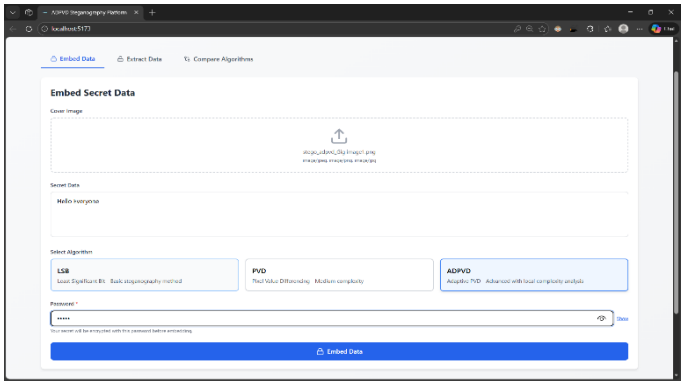


Fig2: Secret Data Embedding Using ADPVD Algorithm: presents the embedding of the secret textual data in the chosen cover image with the help of the ADPVD algorithm. "Hello Everyone" is given as input secret message and the ADPVD technique embeds information into the cover image adaptively depending on the local pixel variations. The figure specifies that the embedding procedure is performed quite effectively without affecting the visual quality of the original image.

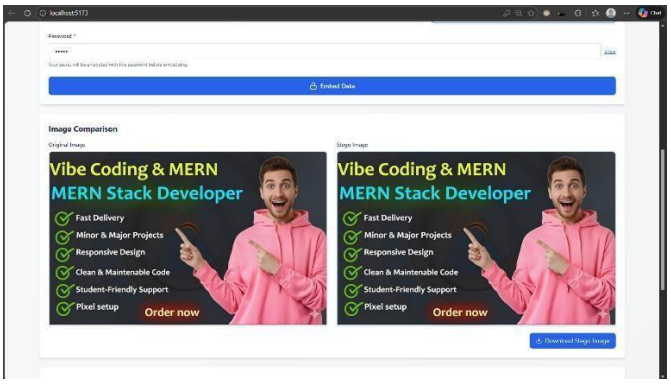


Fig 3: Original Image and Stego-Image Comparison: The following figure presents the visual comparison of the original comparison shows that there is basically no visual difference in both images. That demonstrates the imperceptibility of the cover image and the stego-image obtained after embedding the secret data by using the proposed ADPVD method. The proposed method, since detecting the presence of hidden data is not possible by the human visual system

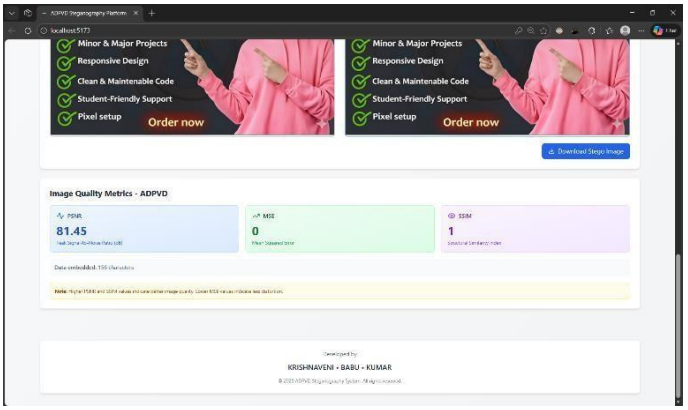


Fig 4: Image Quality Metrics for ADPVD Algorithm: presents the quantitative image quality metrics obtained in applying the ADPVD steganography technique. PSNR, MSE, and SSIM are some of the metrics that have been evaluated. A high PSNR value of 81.45 dB, zero MSE, and an SSIM of 1 provide evidence of superior image quality with the least distortion. These indeed affirm that the ADPVD algorithm is capable of maintaining image integrity

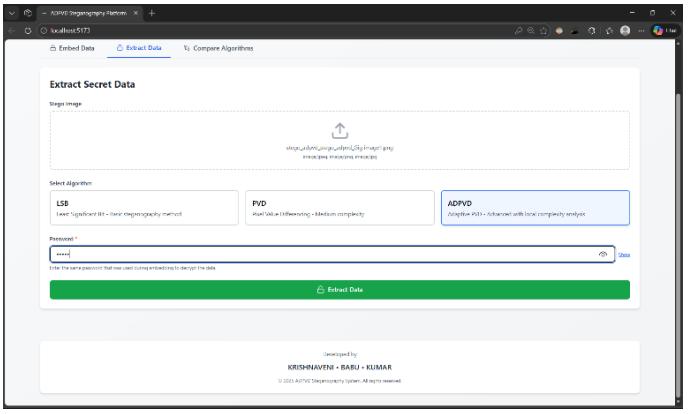


Fig 5: Extract Data Interface of ADPVD Steganography System : This is the extraction module of the proposed steganography system. The user will upload the stego-image, choose the same algorithm that was used during embedding, and enter the correct password for decryption. It protects the process of retrieval of the hidden information from unauthorized persons with password verification and uses an appropriate algorithm for extraction.

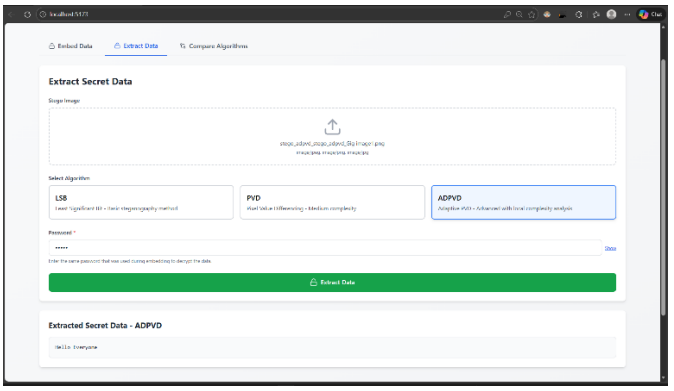


Fig 6: Successful Extraction of Hidden Secret Data: It presents the successful extraction of the hidden secret message from the stego-image via an ADPVD algorithm.

The extracted output completely matched the original input message, which confirmed that the proposed data extraction process is reliable and accurate without the use of any original cover image**.**

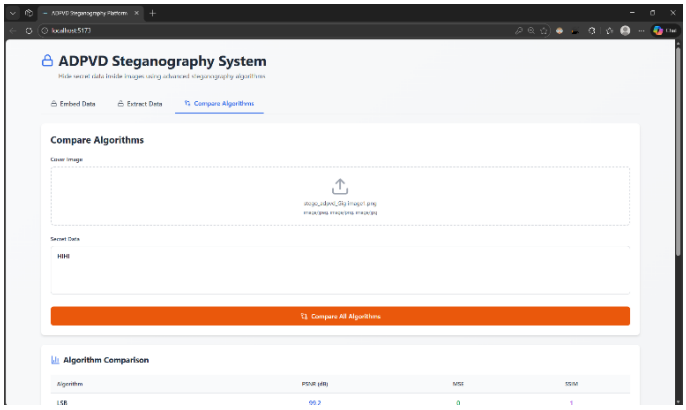


Fig 7: Algorithm Comparison Module Interface: This module compares the different algorithms of the proposed system. Using this module, a user can compare the performances of LSB, PVD, and ADPVD algorithms by using one original cover image with the same secret data. This kind of comparison is helpful in analyzing their strengths and weaknesses under the same condition.

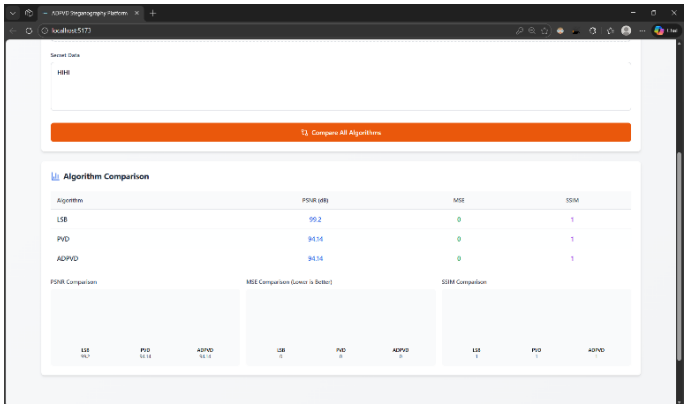


Figure 8: **Comparative Analysis of PSNR, MSE, and SSIM :** Comparison of LSB, PVD, and ADPVD algorithms in terms of PSNR-MSE-SSIM metrics. It can be viewed that the ADPVD gives competitive PSNR values at zero MSE and maximum SSIM for better imperceptibility and structural preservation than traditional techniques. Therefore, this comparison proves the strength of the proposed ADPVD approach in terms of secure image steganography.

CONCLUSION

This paper proposed an efficient image steganography using an Advanced Pixel Value Differentiation approach, which can achieve secure and imperceptible data hiding. The proposed method overcomes the drawbacks of traditional PVD-based techniques by introducing adaptive range selection and optimized pixel pair analysis, enabling improved payload capacity with preservation of high image quality. Experimental

Experimental evaluation indicates that the proposed ADPVD approach yields better performance compared to state-of-the-art methods in terms of imperceptibility, embedding capacity as well as robustness. The high PSNR and SSIM values further point to a better preservation of visual quality of the stego-image for high embedding levels. Moreover, the adaptive embedding provides greater resistance against common steganalysis techniques and image processing- based attacks. In short, the proposed ADPVD-based steganography framework offers a confidential data transmission technique via digital images that is secure, reliable, and computationally efficient [1], [7].

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